Scheme and Answer Key for Valuation- QP Code: 25900565

MGU-UGP (Honours) Regular Examination, October 2025

Third SEMESTER

Discipline Specific Elective (DSE)- MG3DSEPHYA00 - INTRODUCTION TO MEDICAL PHYSICS

Duration: 2 Hours **Maximum Marks: 50**

Part A – Short Answer Type Questions

Answer any 10 questions

Each question carries **2 marks (10x2=20)**

1. Describe the concept of the color triangle in relation to cone excitation.[U, CO1]

   The color triangle represents color perception based on the three cone types in the retina — red (L), green (M), and blue (S). Any visible color can be represented as a combination of the three cone excitations forming a triangular chromaticity diagram.

   2 marks for explaining the three cones and the triangle concept.

2. State the differences between continuous wave (CW) Doppler and pulsed wave (PW) Doppler. [K, CO3]

   CW Doppler: uses continuous transmission and reception; measures high velocities; no range resolution.

   PW Doppler: uses pulses; can measure velocity at specific depth; limited by aliasing.

   1 mark each for difference in operation and limitation.

3. Name any two precordial leads used in ECG. [K, CO6]

   Any two of:

   V1 4th intercostal space, right sternal border

   V2 4th intercostal space, left sternal border

   V3 Midway between V2 and V4

   V4 5th intercostal space, midclavicular line

V5 Same level as V4, anterior axillary line

V6 Same level as V4, midaxillary line

1 mark per correct lead

4. List two uses of photodynamic therapy in eye treatment. [K, CO1]

   Used to treat **(a)** age-related macular degeneration (AMD) and **(b)** choroidal neovascularization.

   1 mark each for two correct uses.

5. State the formula for acoustic impedance in terms of density and speed of sound. [K, CO3]

   $Z=\rho c$, where Z = acoustic impedance, ρ = density of medium, c = speed of sound.

   1 mark for formula, **1 mark** for explanation.

6. Explain why scotopic (night) vision images appear bluish-green. [U, CO1]

   Scotopic vision uses **rod cells** which are most sensitive to **wavelengths around 507 nm (blue-green region)**; hence night vision appears bluish-green.

   2 marks for reason linked to rods and wavelength.

7. Describe two clinical uses of speech audiometry. [K, CO4]

   1)Validation of Pure Tone Thresholds (Speech Recognition Threshold - SRT)
   2) Differential Diagnosis (Word Recognition Score - WRS)

   1 mark each for correct use.

8. Explain the function of the eustachian tube in hearing. [U, CO4]

   Equalizes air pressure on both sides of the tympanic membrane; connects middle ear to nasopharynx; ensures optimal vibration of eardrum.

   2 marks for full explanation.

9. List the different regions of the electromagnetic spectrum in order of increasing frequency. [K, CO1]

Radio waves → Microwaves → Infrared → Visible → Ultraviolet → X-rays → Gamma rays.

2 marks for correct order.

10. Explain the parts of a transducer. [U, CO5]

Transducers consist of two basic parts which are:

- **The Sensor part** senses a particular type of energy. It gives a response according to the physical phenomenon that it senses. The response serves as the input for the transducing element.
- **The Transducing part** converts the energy sensed by the sensor into some other form of energy.

1 marks for each part and function.

11. Define an endoscope and mention its basic function. [K, CO2]

-An Endoscope is a medical tool used to examine organs. It consists of a soft long tube with a small light source and camera at one end which can transmit images to a display.
-Its basic function is to allow a doctor to look inside the body without major surgery by inserting it through a natural opening, such as the mouth or anus. The camera projects images onto a screen, enabling the physician to diagnose

1 mark for definition, **1 mark** for function.

12. Explain briefly EEG. [K, CO6]

-An EEG, or electroencephalogram, is a non-invasive medical test that measures and records the brain's electrical activity through electrodes placed on the scalp.
-It is used to diagnose neurological conditions like epilepsy, sleep disorders, and brain injuries.

1 mark for definition, **1 mark** for use.

13. Define frequency of sound wave and state its SI unit. [K, CO3]

Frequency = number of cycles per second of sound wave. Sound waves with frequencies in the range 20 Hz to 20 kHz can be detected by the human ear

SI unit - hertz (Hz)**.**

1 mark for definition, **1 mark** for unit.

14. What is the principle of CT imaging? [K, CO6]

Computed Tomography (CT) Scan is a non-invasive medical imaging procedure that utilizes a combination of X-rays and sophisticated computer technology to create detailed cross-sectional images

The fundamental principle of a CT scan relies on the differential **attenuation** (absorption or scattering) of X-ray beams as they pass through different types of tissue in the body.

1 mark each for describing CT and principle.

Part B – Short Essay Type Questions

Answer any 6 questions
Each question carries 5 marks (6x5=30)

15. Explain the function of the interferometer in OCT. [K, CO2]

Optical Coherence Tomography (OCT) is a non-invasive, non-contact imaging technique that uses low-coherence near-infrared light and interferometry **to** obtain high-resolution, cross-sectional images of biological tissues, particularly the retina and optic nerve in ophthalmology.

Role of the Interferometer in OCT

-Splitting the light: The interferometer (usually a Michelson interferometer**)** splits this light into two beams:Sample arm → directed into the tissue (eye), Reference arm → directed onto a mirror with a known position

-Collecting reflections: Light reflects back from different layers of the tissue (retina, cornea, etc.) with varying intensities. At the same time, the reference mirror reflects light back along the reference arm.

-Producing interference:The interferometer recombines the sample beam and reference beam**.** Because the light source is *low-coherence,* interference occurs only when the path length of the sample reflection matches the reference arm within the coherence length**.** This interference encodes depth information

-Generating A-scan:By analyzing the interference signal, the OCT system produces an **axial reflectivity profile (A-scan)** of the tissue. Multiple A-scans arranged side by side give a **B-scan (cross-sectional image)**.

2 mark– OCT definition, 3 mark– role of interference in OCT

16. Explain the mechanism of light dispersion with the help of a prism. [U, CO1]

    Dispersion: separation of white light into constituent colors due to **variation of refractive index with wavelength**.
    • Prism refracts each wavelength at different angles; shorter wavelengths (violet) bend more than longer (red).
    • Formula: n=[sin(A+D)/2]/sin(A/2)
    • Results in spectrum (VIBGYOR).

    1 – Definition, 2 – Mechanism, 1 – Example (VIBGYOR), 1 – Diagram/expression

17. Describe the input characteristics of a transistor in CB mode. [K, CO5]

    In Common Base (CB) configuration, input = emitter–base circuit, output = collector–base circuit.
    • Input characteristics: plot of **emitter current (I_E)** vs **emitter–base voltage (V_{EB})** for constant collector–base voltage (V_{CB}).
    • Shows exponential rise similar to diode.
    • Used to determine input resistance (low).

    1mark – Circuit diagram, 2 mark– Explanation, 1 mark– Input characteristics plot, 1mark – input resistance

18. Differentiate between CO_2, Nd:YAG, and excimer lasers in terms of wavelength and applications. [U, CO1]

    CO_2 Laser: λ = 10.6 μm; infrared; used in surgery (cutting, cauterizing soft tissue).
    Nd:YAG Laser: λ = 1064 nm; near-infrared; used in ophthalmology, dermatology, dentistry.
    Excimer Laser: $\lambda \approx$ 193–351 nm; UV; used in corneal reshaping (LASIK).

    2marks – Principle of each laser, 1.5 marks – Any one applications of each laser, 1.5 marks – Wavelength range of each laser

19. Describe briefly the process of attenuation of absorption. [K, CO3]

    -When an ultrasound wave propagates through soft tissue, the energy associated with the wave is gradually lost so that its intensity reduces with distance travelled, an effect known as attenuation

-Caused by absorption**,** scattering**,** and reflection**.**
-The most important mechanism is absorption, in which ultrasound energy is converted into heat.

-At low frequencies, the particles are able to move in step with the passing pressure wave and energy associated with the motion of the particles is effectively passed back to the wave as it moves on.

-Absorption has been found to be strongly dependent on tissue composition and structure.

-Tissues with high collagen content such as tendons and cartilage show high absorption, whereas those with high water content show lower absorption.

=Water and liquids such as urine, amniotic fluid and blood have low absorption and low attenuation.

1 mark– Attenuation definition, 2 marks – atteneuation of absorption, 2 marks– Examples

20. What are the properties of X-rays that make them useful in medical imaging? [K, CO5]

    - High energy, short wavelength (0.01–10 nm).
    - Penetrate soft tissues but absorbed by dense materials (bone).
    - Cause ionization and fluorescence.
    - Travel in straight lines.
    - Can be recorded on film/detectors to produce contrast images.

    Any five properties (1 mark each)

21. Why are op-amps widely used in biomedical instruments? [U, CO5]

    -High input impedance and low output impedance → minimal loading on sensors.
    -High gain → amplify weak physiological signals (e.g., ECG, EEG).
    =Stable and low noise.
    -Can be configured as filters, amplifiers, integrators, or differentiators.
    -Compact, reliable, and inexpensive.

    -Applications (Ellaborate any three biomedical applications)

    2 marks– Op amp properties, 3marks– Biomedical applications

22. Describe the functioning of electromagnetic tracking systems in freehand ultrasound. [K, CO4]

- Small transmit device placed near the patient generates a magnetic field.
- A sensor coil is attached to the ultrasound transducer, and is connected to a receiver.
- Movement of the coil through the magnetic field induces voltage in the coil from which the receiver estimates the position and orientation in space.
- These systems do not require line of sight (i.e. a clear uninterrupted path between coil and receiver); however, they may suffer distortions due to the presence of other hospital equipment

2 marks– Components, 2 marks– Working, 1 mark– Application